

Roll No.

TPE-002

B. TECH. (ECE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
OPTICAL FIBER COMMUNICATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Draw and explain the general block diagram of optical fiber communication. 10 Marks (CO1)

OR

- (b) Discuss the advantages of optical fiber communication over other means of communication. 10 Marks (CO1)

2. (a) Briefly describe the scattering losses occurring in optical fiber. 10 Marks (CO2)

OR

- (b) Discuss absorption losses in optical fibers, by comparing and contrasting the intrinsic and extrinsic absorption mechanisms.

10 Marks (CO2)

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3. (a) Describe the following with the aid of simple ray diagrams :

10 Marks (CO1)

- (i) Multimode step index fiber
- (ii) Single mode step index fiber

OR

- (b) The refractive indices of core and cladding materials of a step index fiber are 1.48 and 1.45 respectively. Calculate :

10 Marks (CO1)

- (i) numerical aperture
- (ii) acceptance angle
- (iii) the critical angle

4. (a) Explain mode field diameter and spot size.

10 Marks (CO2)

OR

- (b) When a long single mode fiber is operating at a wavelength of 1.3 μm , the attenuation is 0.5 dB/km. The core diameter of the fiber is 6 μm and the laser source bandwidth is 700 MHz. Calculate the threshold optical powers for stimulated Brillouin and Raman scattering. 10 Marks (CO2)

5. (a) Briefly explain the reason for pulse broadening due to material dispersion in optical fibers.

10 Marks (CO1 & CO2)

OR

- (b) A step index fiber in air has a numerical aperture of 0.16, a core refractive index of 1.45 and a core diameter of 60 μm . Determine the normalized frequency for the fiber when light wavelength of 0.9 μm is transmitted.

10 Marks (CO1 & CO2)